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RE: 2024 AIAA Undergraduate Scholarship

Dear selection committee.

First and foremost, I write many letters of recommendation for many students, with no student being as impressive as Jacob Vaught on so many levels. Jacob is an integral aspect of the College of Engineering and Computing here at the University of South Carolina, being active in clubs, activities, and research everywhere I turn. He is always willing to help, guiding others to research opportunities, and in general supporting anyone he can. This trait has been noticed by many professors and everyone speaks very highly of him. It goes without saying that I am enthusiastically recommending Jacob Vaught for a South Carolina Space Grant Undergraduate Research Award; both on his technical merit but also as a way to support a student that will soon be a national and international leader in the aerospace sector.

Jacob started working with my group in the spring of 2022, while also doing research with other groups. During that time, he has worked on the deployment of FEA models to real-time computers and the FEA-based modeling of structures. His work has been integral to the lab and is the co-author of a published conference paper that was recently presented at IMAC titled "Modal analysis using a UAV-deployable wireless sensor network". Moreover, he is also a co-author of another submitted paper from our group. Recently, Jacob has switched projects and is now developing Field Programmable Gate Arrays (FPGAs) based machine learning accelerators to enable real-time digital twins of power electronics for electric aircraft.

While his resume will certainly speak for itself, I want to again reiterate how helpful Jacob is around the department while also being an extremely productive researcher and working as my lab manager. He is truly deserving of a South Carolina Space Grant Undergraduate Research Award.

If you have any questions, please do not hesitate to contact me.

Sincerely,

Austin Downey